Diversity

What is it?

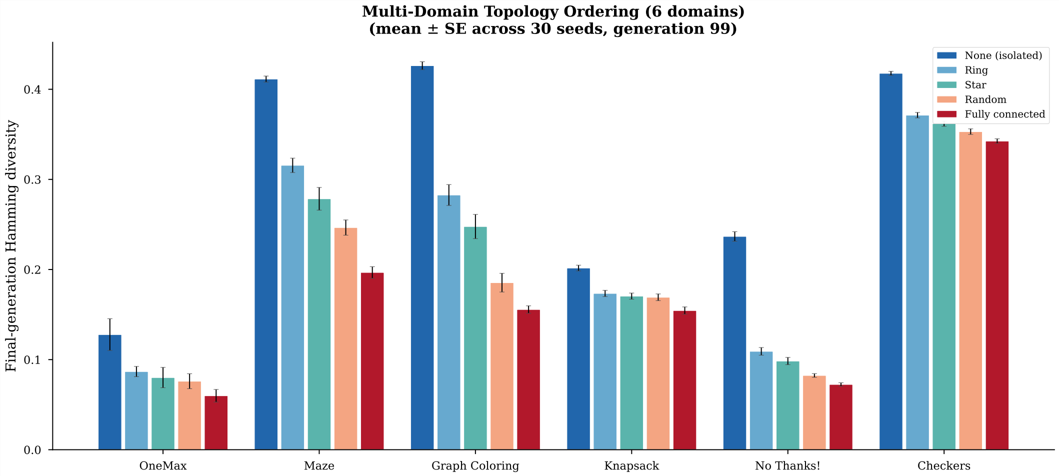
How different are the individuals in your population?

Measured as mean pairwise Hamming distance (normalized 0–1).

Why do we care?

- Too little $\rightarrow$ premature convergence, stuck in local optima
- Too much $\rightarrow$ no selection pressure, random search
- The sweet spot depends on the problem — but what controls it?

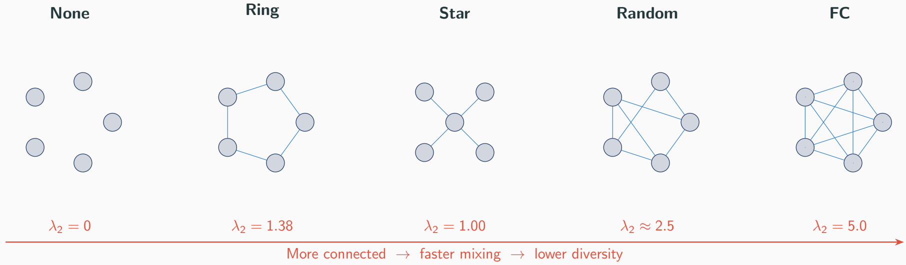
Multi-Domain Topology Ordering



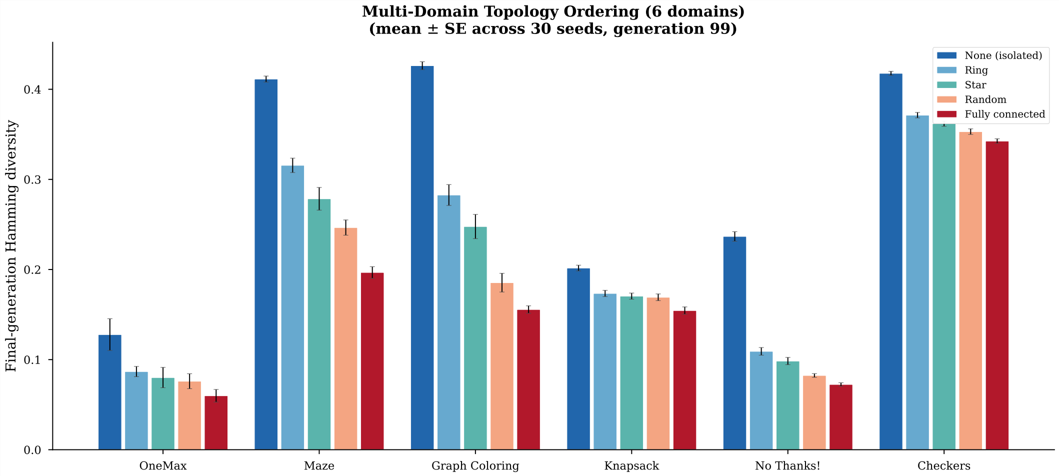
The Domains

1. OneMax
2. Maze (breed difficult perfect mazes — spanning trees on a grid — via edge-swap mutation & union crossover)
3. Graph Coloring
4. Knapsack
5. No Thanks!
6. Checkers

Island Models & Topology



Multi-Domain Topology Ordering



Statistics

Kendall's W

$W = 1.0$

Perfect concordance

$\chi^2 = 24.0, \quad p = 0.00008$

Spearman

All 15 pairwise $\rho = 1.0$

Two-way ANOVA

$F_{\text{topo}} = 47.8, \quad p < 10^{-6}$

$F_{\text{domain}} = 0.13, \quad p = 0.945$

Variance ratio

$23.9\times$

Topology $\gg$ domain

The End

Nobody likes a long talk,
so you can tune out here if you like.

Up next: how Haskell and category theory
led us to this result.

Kleisli Arrows

Every GA operator has the same type:

$$\texttt{Pop}\ a \longrightarrow M\,(\texttt{Pop}\ a)$$

Population in, population out, effects bundled in M:

```
type EvoM = ReaderT GAConfig
              (WriterT GALog
                (State StdGen))
```

Kleisli composition ($\ggg$) chains them:

```
pipeline = evaluate >=> select
                    >=> crossover
```

Why?

All code to reproduce these experiments:

`https://github.com/lyra-claude/categorical-evolution`